

点面关系

平面的点法式方程

平面过点 $M_0(x_0, y_0, z_0)$ 法向量 $n = (A, B, C)$ 则平面方程为

$$A(x-x_0) + B(y-y_0) + C(z-z_0) = 0$$

平面一般式方程

$$\text{令 } D = -Ax_0 - By_0 - Cz_0$$

$$\text{有 } Ax + By + Cz + D = 0$$

平面三点式方程

$$M_1(x_1, y_1, z_1), M_2(x_2, y_2, z_2), M_3(x_3, y_3, z_3)$$

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

平面截距式方程

$$M_1(a, 0, 0) \quad M_2(0, b, 0), \quad M_3(0, 0, c)$$

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

过原点 $Ax + By + Cz = 0$

平行于x轴 $By + Cz + D = 0$

过y轴 $By + Cz = 0$

平行于xOy $Cz + D = 0$

点到平面的距离 (点 $M_1(x_1, y_1, z_1)$ 到平面 $Ax + By + Cz + D = 0$)

$$d = |\overrightarrow{M_0 M_1} \cdot \vec{n}|$$

$$\Rightarrow d = |(x_1 - x_0, y_1 - y_0, z_1 - z_0) \cdot \frac{(A, B, C)}{\sqrt{A^2 + B^2 + C^2}}|$$

$$\Rightarrow d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

面面关系

$$\pi_1: Ax + By + Cz + D_1 = 0$$

$$\pi_2: Ax + By + Cz + D_2 = 0$$

平行 $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2} \neq \frac{D_1}{D_2}$

垂直 $A_1 A_2 + B_1 B_2 + C_1 C_2 = 0$

夹角 $\cos \varphi = |\cos \theta| = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| \cdot |\vec{n}_2|} = \frac{|A_1 A_2 + B_1 B_2 + C_1 C_2|}{\sqrt{A_1^2 + B_1^2 + C_1^2} \cdot \sqrt{A_2^2 + B_2^2 + C_2^2}}$

直线方程

直线的点向式方程、标准式方程、对称式方程

$$\frac{x-x_0}{l} = \frac{y-y_0}{m} = \frac{z-z_0}{n} = t \quad \text{化为平面交线理解}$$

直线的参数方程

过点 $M_0(x_0, y_0, z_0)$ $S = (l, m, n)$

$$x = x_0 + lt, \quad y = y_0 + mt, \quad z = z_0 + nt \quad -\infty < t < +\infty$$

直线的两点式方程

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{y_2-y_1} \quad \text{化为点向式方程理解}$$

直线的一般式方程

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0 \\ A_2x + B_2y + C_2z + D_2 = 0 \end{cases}$$

直线外一点到直线的距离

$$d = \frac{|\vec{M_0M_1} \times \vec{S}|}{|\vec{S}|}$$

例: $M_1(1, 2, 3)$ 到直线 $L: \begin{cases} x - y + z + 5 = 0 \\ 5x - 8y + 4z + 36 = 0 \end{cases}$ 的距离

$$S = n_1 \times n_2 = (1, -1, 1) \times (5, -8, 4) = (4, 1, -3)$$

$$\text{令 } x=0 \text{ 解得 } y=4, z=-1 \Rightarrow \overrightarrow{M_0 M_1} = (1, -2, 4)$$

$$d = \frac{|(1, -2, 4) \times (4, 1, -3)|}{|(4, 1, -3)|} = \frac{|(2, 19, 9)|}{|(4, 1, -3)|} = \sqrt{\frac{333}{13}}$$

直线的关系

直线共面

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad (L_1) \quad P_1(x_1, y_1, z_1)$$

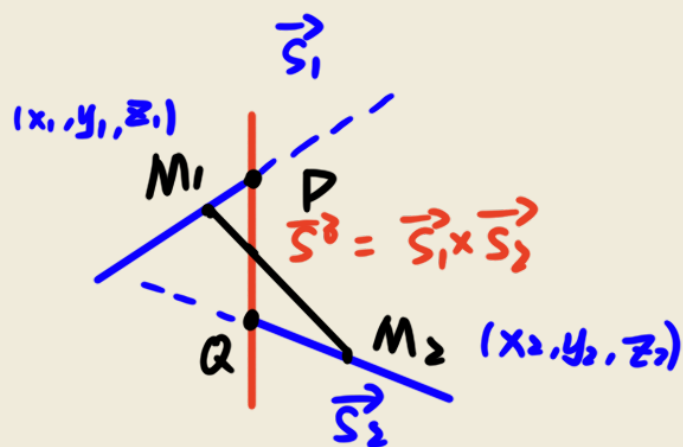
$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad (L_2) \quad P_2(x_2, y_2, z_2)$$

$$L_1, L_2 \text{ 共面} \Leftrightarrow \begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad \text{混合积为0} \quad \overrightarrow{P_1 P_2} (\vec{S}_1 \times \vec{S}_2) \quad \text{平行六面体体积}$$

⇐ 即 L_1, L_2 的法向量垂直于面

异面直线公垂线

$$\textcircled{1} \text{ 先求方向 } \Rightarrow \vec{S} = \vec{S}_1 \times \vec{S}_2$$



$$\vec{S}(x, y, z)$$

② 再求长度

$$|\vec{PQ}| = |\vec{PQ} \cdot \vec{S}_0| = |(\vec{PM}_1 + \vec{M_1M_2} + \vec{M_2Q}) \cdot \vec{S}_0| = |\vec{M_1M_2} \cdot \vec{S}_0|$$

$$= \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}$$

③ 求公垂线

π_1 : L 与 L_1 决定的平面

$$n_1 \cdot \vec{n}_1 = \vec{S}_0 \cdot \vec{S}_1$$

π_2 : L 与 L_2 决定的平面

$$n_2 \cdot \vec{n}_2 = \vec{S}_0 \cdot \vec{S}_2$$

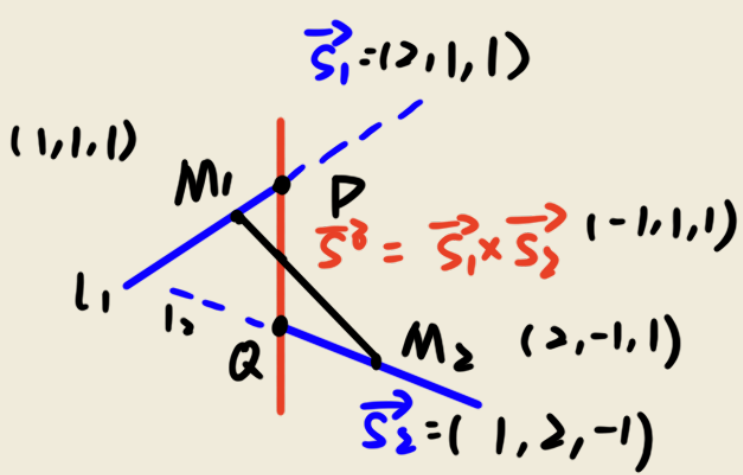
法一 $L = \begin{cases} \pi_1 \\ \pi_2 \end{cases}$

法二 $\vec{S} \cap Q = \begin{cases} \pi_1 \\ L_2 \end{cases}$

法三 $\begin{cases} \overrightarrow{M_1(t)M_2(s)} \cdot \vec{S}_1 = 0 \\ \overrightarrow{M_1(t)M_2(s)} \cdot \vec{S}_2 = 0 \end{cases}$

例. 求 $L_1: \frac{x-1}{2} = y-1 = z-1$ $L_2: x-2 = \frac{y+1}{2} = \frac{z-1}{-1}$ 距离. 公垂线方程

$$\vec{S}_0 = \vec{S}_1 \times \vec{S}_2 = (-3, 3, 3)$$



$$d = |\overrightarrow{PQ} \cdot \vec{S^0}| = |\overrightarrow{M_1 M_2} \cdot \vec{S^0}| = \left| \frac{(1, -2, 0) \cdot (-1, 1, 1)}{\sqrt{3}} \right| = \sqrt{3}$$

法一:

$$\pi_1 = \begin{vmatrix} x-1 & y-1 & z-1 \\ 1 & -1 & -1 \\ 2 & 1 & 1 \end{vmatrix} = 0 \Leftrightarrow -(y-1) + (z+1) = 0 \Leftrightarrow y-z=0$$

$$\pi_2 = \begin{vmatrix} x-2 & y+1 & z-1 \\ 1 & -1 & -1 \\ 1 & 2 & -1 \end{vmatrix} = 0 \Leftrightarrow (x-2) + (z-1) = 0 \Leftrightarrow x+z-3=0$$

$$\Rightarrow L \quad \begin{cases} y-z=0 \\ x+z-3=0 \end{cases}$$

法二:

求 π_1 与 l_2 交点 Q

$$\begin{cases} y-z=0 \\ \frac{x-2}{1} = \frac{y+1}{-1} = \frac{z-1}{-1} \end{cases} \Rightarrow Q\left(\frac{8}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$\Rightarrow L \quad \frac{x-\frac{8}{3}}{1} = \frac{y-\frac{1}{3}}{-1} = \frac{z-\frac{1}{3}}{-1}$$

法三: 直接求点, 不求 π_1, π_2

$$L_1: x=1+2t, y=1+t, z=1+t$$

$$L_2: x=2+u, y=-1+2u, z=1-u$$

$$\text{交点 } P, Q \quad \overrightarrow{PQ} \parallel \vec{S} \Rightarrow \frac{1+u-2t}{1} = \frac{-2+2u-t}{1} = \frac{-u-t}{-1}$$

$$\Rightarrow \begin{cases} 3u-3t=1 \\ 3u-2=0 \end{cases} \Rightarrow \begin{cases} u=\frac{2}{3} \\ t=\frac{1}{3} \end{cases}$$

$$P\left(\frac{5}{3}, \frac{4}{3}, \frac{4}{3}\right) \quad Q\left(\frac{8}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

投影平面

例 $L: \frac{x+1}{4} = \frac{y-2}{-1} = \frac{z-1}{5}$ 平面 π 的方程为 $3x+y+2z+20=0$. 求直线 L 与 π 的夹角和交点 M . L 在 π 上的投影平面与投影方程. (点法式计算)

方向向量 $S=(4,-1,5)$ $\pi: n=(3,1,2)$

$$\sin \varphi = \frac{4 \cdot 3 + (-1) \cdot 1 + 5 \cdot 2}{\sqrt{4^2 + (-1)^2 + 5^2} \cdot \sqrt{3^2 + 1^2 + 2^2}} = \frac{\sqrt{3}}{2} \Rightarrow \varphi = \frac{\pi}{3}$$

$L: x=-1+4t, y=2-t, z=1+5t$ 代入 π

$$\Rightarrow 3(-1+4t) + (2-t) + 2(1+5t) + 20 = 0 \Rightarrow t = -1$$

$\Rightarrow$ 交点 M 坐标 $(-5, 3, -4)$

π_1 法向量 n_1

$$S \times n = (4, -1, 5) \times (3, 1, 2) = -7(1, -1, -1)$$

投影平面 $\pi_1: (x+5) - (y-3) - (z+4) = 0 \Rightarrow x - y - z + 4 = 0$

$$L \text{ 在 } \pi \text{ 上的投影 } \begin{cases} 3x+y+2z+20=0 \\ x-y-z+4=0 \end{cases}$$

平面束

$$\begin{cases} \pi_1: A_1x + B_1y + C_1z + D_1 = 0 \\ \pi_2: A_2x + B_2y + C_2z + D_2 = 0 \end{cases}$$

$$\text{平面束方程 } \pi_\lambda: A_1x + B_1y + C_1z + D_1 + \lambda(A_2x + B_2y + C_2z + D_2) = 0$$

重写“例”

$$L: \begin{cases} x+4y-7=0 \\ 5y+z-11=0 \end{cases} \quad \text{投影平面方程 } \pi_\lambda: x+4y-7+\lambda(5y+z-11)=0$$

$$n_\lambda = (1, 4+5\lambda, \lambda)$$

$$n_\lambda \cdot n = 0 \Rightarrow 1 \cdot 3 + (4+5\lambda) \cdot 1 + \lambda \cdot 2 = 0 \Rightarrow \lambda = -1$$

$$\Rightarrow \text{投影平面方程 } x-y-z+4=0$$